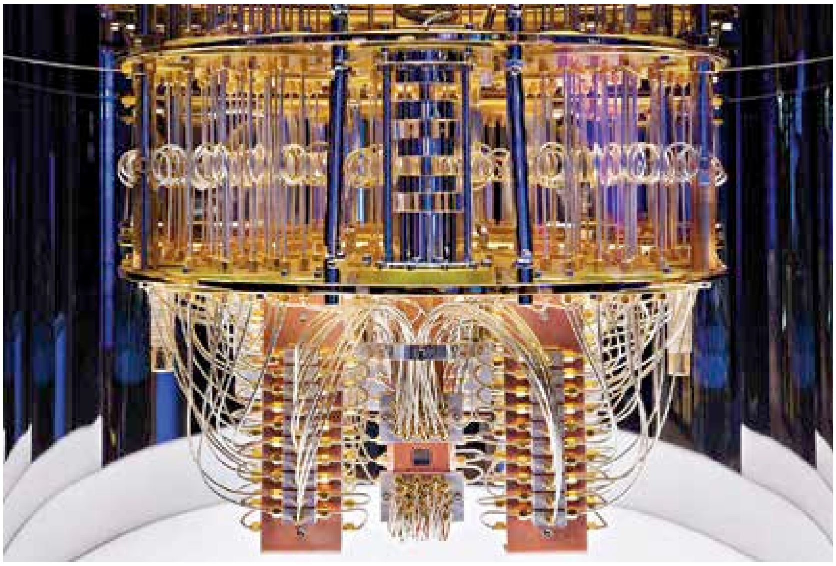
density would only be possible via such stacking methods. So while Moore's Law might not hold true in the 2D space, it should last for a lot longer in the 3D space. And once those methods become stale, you'll have to drastically reimagine the way we design processors. That's when binary would fall short and the era of computers using more than two states would become mainstream.

## QUANTUM COMPUTING

We say "mainstream" because quantum computers are already here. The concept of quantum computers and their theorised potential has been talked over in research circles and mainstream publications for so long that one would assume that quantum computers are commonplace. The reality is that quantum computers have not exited the laboratory. The fundamental unit of computing for quantum computers i.e. qubits, are so unstable that they have to be maintained in supercooled environments to be operational. And even in that state, they haven't achieved quantum supremacy.

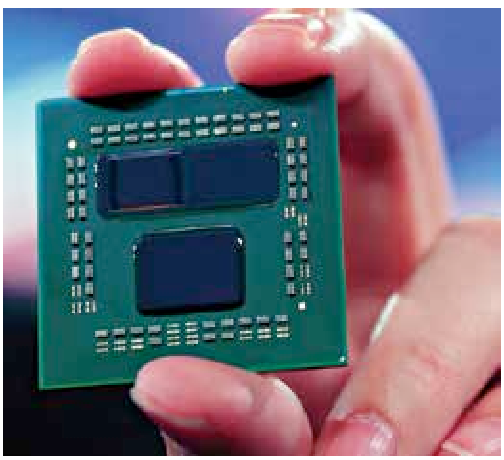
Quantum supremacy is the threshold of computing power that quantum computers need to achieve in order to be deemed better than traditional computers. Google claims to have achieved quantum supremacy on their Sycamore processor which was developed along with NASA, but it's competitor IBM was quick to refute those claims. IBM stated that in the strictest definition of quantum supremacy, Google' Sycamore does not meet the requirements and that fundamentally, quantum computers will never reign over traditional computers and will have to work in concert with them.



Credit: IBM Research

Quantum Computer at IBM

While the two technology giants fight over definitions, the goal of miniaturising quantum computers is yet to be solved. Intel did showcase a small fingernail-sized quantum computer but it's in no way close to achieving quantum supremacy. Their target is to achieve production-level quantum computers in the next 10 years. By all means, this would be akin to what IBM stated. A symbiotic relationship formed between a quantum computing chip with a traditional processor. Much like how accelerators are produced. The only thing is that in the next 10 years, we're going to see enterprise level applications because no one in the world will be able to package that in a form-factor that appeals to everyday consumers. That will take another 7-10 years or more.



AMD 3D V-Cache Prototype CPU

## ARM AND OPEN COMPUTING

ARM is the future of portable computing. It is ubiquitous and powers every portable smartphone in existence. And Snapdragon's 8CX platform is slowly inching towards becoming viable enough to start replacing the low-performance notebooks. In the next 10 years, we will have ARM processors maturing to the extent that your laptop and ours would be based on ARM. ARM processors are already being used to make supercomputers so the whole notion of ARM being a poor performer compared to x86 will soon go away.

Then there's the concept of open architectures. RISC-V is steadily gaining a strong foothold in the custom silicon business. With an open architecture, companies don't have to pay ridiculous royalties and the cost of scaling production goes down considerably. However, RISC-V isn't as organised as one would hope for mass adoption into the consumer space. We hope that will change in the coming decade.

## SO WILL THE PC STAY?

Most certainly. And we're talking about desktops as well. While smartphones have become capable enough to displace the PC, there will always be a use case that requires sheer computing power that a portable device can never provide. Analysts and publications have been ringing the death knell for PC for more than a decade and we're yet to see a downward trend in the PC space. So yes, the PC is here to stay but the form-factor is destined to grow smaller and smaller.